TIC TAC TOE GAME

A MINI PROJECT REPORT

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ABSTRACT

The Tic Tac Toe Game is a console-based mini project developed using C++. The project is designed to demonstrate fundamental programming concepts such as arrays, loops, functions, and conditional statements. It allows two players to play the game interactively on a 3×3 board. The game dynamically updates the board

after every move and checks for winning or draw conditions. Replay functionality is also included to improve user interaction and gameplay experience.

OBJECTIVES

- To develop a console-based Tic Tac Toe game using C++.
- To understand the use of arrays for storing game data.
- To implement loops and conditional statements for game logic.
- To create an interactive two-player gaming application.
- To implement win, loss, and draw detection mechanisms.
- To improve problem-solving and logical thinking skills.

FEATURES

- Interactive two-player gameplay.
- Dynamic display of the game board after each move.
- Input validation for correct player moves.

- Win detection for rows, columns, and diagonals.
- Draw detection when all cells are filled.
- Replay option to restart the game without exiting.
- Simple and user-friendly console interface.

ALGORITHM

1. Start the program.

2. Initialize the 3×3 game board with numbers 1–9.

3. Set the first player as Player X.

4. Display the game board.
5. Ask the current player to enter a position.
6. Check whether the entered position is valid.
7. Update the board with the player's symbol.
8. Check for a winning condition:
    a. Same symbols in a row

    b. Same symbols in a column

    c. Same symbols diagonally

9. If a player wins, display the winner and end the game.
10. If all positions are filled without a winner, declare the game as a draw.

11. Ask the user whether to replay the game.
12.If yes, restart the game; otherwise, exit the program.
13.Stop the program

OUTPUT

INITIAL BOARD

```
 1 | 2 | 3
---|---|---
 4 | 5 | 6
---|---|---
 7 | 8 | 9

Player X, enter position (1-9):
```

AFTER FIRST MOVE

```
 X | 2 | 3
---|---|---
 4 | 5 | 6
---|---|---
 7 | 8 | 9

Player O, enter position (1-9):
```

MID GAME

```
 X | X | 3
---|---|---
 4 | O | 6
---|---|---
 7 | O | 9

Player X, enter position (1-9):
```

WINNING OUTPUT

```
 X | X | X
---|---|---
 4 | O | 6
---|---|---
 7 | O | 9

Player X wins!

Play again? (y/n):
```

EXIT MESSAGE

```
Thank you for playing!
```

CONCLUSION

The Tic Tac Toe Game project successfully demonstrates the implementation of core C++ programming concepts in a simple and interactive manner. The project uses arrays, loops, functions, and conditional logic effectively to manage game operations. It provides an engaging way to understand game development basics and enhances programming and logical thinking skills.